

ADITYA KULKARNI

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SUMMARY

Full-stack engineer (MS CS, NC State, May 2026) shipping features end-to-end, from **PostgreSQL** schema to **React.js** UI. Delivered **10+ production features** on a multi-tenant insurance platform and built the **LLM-agent** infrastructure behind them.

EXPERIENCE

NC State University <i>Graduate Research Assistant</i>	Apr 2025 – Present Raleigh, NC
- Designed a cloud-based genomic platform (JBrowse 2 on Google Cloud Run) enabling 100+ researchers to interactively explore 500GB of strawberry and grape datasets - Collaborated on ML architecture design, applying hyperparameter tuning and time-series analysis to improve model accuracy by 12%	
Isomer AI <i>Product Engineer Intern</i>	May 2025 – May 2026 Remote
- Shipped 10+ features end-to-end on a per-tenant insurance SaaS; owned Auth0 RBAC, Redis caching, and PostgreSQL data modeling across isolated customer stacks - Built MCP servers in Python and TypeScript orchestrating 3 LLM vendors (Claude, GPT-4o, Gemini) behind typed extraction schemas, with Playwright browser automation - Engineered a Temporal.io workflow engine with versioned Draft/Live/Archived releases, canary splits, and risk detection; shipped via Docker + GitHub Actions CI/CD with Sentry	
Mphasis (HP Company) <i>Software Developer Intern</i>	Jun 2024 – Aug 2024 Bangalore, India
- Built 5+ AI workflows, including a health-claim automation tool using OCR, NLP, and ML classification, that improved automated claim accuracy by 25% - Automated deployments on AWS and GCP with Grafana monitoring dashboards, and supported model validation and red-teaming to enhance production reliability	
Capgemini <i>AI/ML Intern</i>	Jun 2023 – Aug 2023 Mumbai, India
Led the SDLC for a support-ticket automation system, cutting manual processing time 15%; built Hadoop, Spark, and Kafka pipelines that reduced processing latency 20%	

PROJECTS

PulseRank <i>Real-Time Leaderboard & Activity Feed</i>	2026
- Architected an event-driven leaderboard with sub-millisecond reads via Redis sorted sets, a MySQL outbox for durable writes, and live updates over Server-Sent Events - Built end-to-end in TypeScript, NestJS, and Next.js; verified multi-store consistency with Testcontainers and load-tested throughput with k6	
InventoryReservationService <i>Transactional REST API</i>	2026
- Engineered a Java 21 + Spring Boot inventory API preventing overselling under high concurrency via PostgreSQL row-level locking (SELECT FOR UPDATE) and idempotent endpoints - Hardened with Flyway migrations, Spring Security RBAC, RFC 9457 errors, Testcontainers concurrency tests, and a Dockerized GitHub Actions CI pipeline	
AlgoPulse <i>Agentic AI Platform</i>	2025
- Diagnosed an API-polling bottleneck and re-architected to Supabase Edge Functions that stream LLM reasoning loops directly to the client at 2,000 async req/sec - Integrated Temporal.io durable workflows and Kafka event streaming for fault-tolerant, replayable multi-step agent runs	

EDUCATION

NC State University <i>M.S. Computer Science</i>	Aug 2024 – May 2026 Raleigh, NC
Relevant Coursework: Software Engineering, Operating Systems, Algorithms, Object-Oriented Development, Machine Learning, Data Science	
Mumbai University <i>B.Tech Computer Science, CGPA 9.32/10</i>	Aug 2020 – May 2024 Mumbai, India
Relevant Coursework: Data Structures & Algorithms, Data Warehousing, Big Data Infrastructure, Blockchain, Software Engineering	

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, Ruby, SQL
Frameworks: React.js, Next.js, NestJS, Express.js, Node.js, Spring Boot, FastAPI
Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase, Elasticsearch
DevOps & Tools: Docker, Kubernetes, AWS, GCP, GitHub Actions, Temporal.io, Kafka, Playwright, Testcontainers, Grafana, ArgoCD